# Planners Exploit Deterministic Error: Dithered Low-Bit World Models

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## Abstract

Sampling-based planners query a world model thousands of times per decision, yet low-bit world models fail at planning in ways that bit-width and one-step error do not explain. We show that what matters is the structure of the rounding error rather than its size. Deterministic round-to-nearest leaves an error field that persists across optimizer iterations; in the four tasks where we tested the mechanism, an iterating planner drifts toward the candidates that this error favors. Replacing round-to-nearest with dither (stochastic rounding) shared by all candidates and redrawn at every planner iteration removes most of this drift, at the same bit-width and without training. In internally pre-registered experiments with TD-MPC2 (MPPI, 12 tasks) and DINO-WM (CEM, 3 tasks), deterministic rounding cost up to 100 points of success although its one-step error was no larger than dither's in 29 of 36 affected cells. Dither removed at least half of the damage in all 19 affected Meta-World cells (median 95 percent) and in 4 of 5 DINO-WM cells; in a fused integer kernel (TD-MPC2) the median extra latency was under 10 percent. The fix requires an iterating planner and helps less where little favorable error persists, as on DMC.

***Keywords***— *World models, model-predictive control, quantization, stochastic rounding, robot planning*

### *A. Problem Setup*

An MPC planner queries its world model candidates $\times$ iterations $\times$ horizon times per decision: $512 \times 6 \times 3$ for the MPPI of TD-MPC2 [1] and $300 \times 30 \times 5$ for the CEM of DINO-WM on PushT [2]. Low-bit inference is the most direct way to cut this cost, but quantized world models break planning in ways that bit-width and offline error do not explain [3, 4]. We quantize hidden MLP activations (four dynamics and reward layers in TD-MPC2, six predictor blocks in DINO-WM) to $k$ bits on a calibrated uniform grid. Per task, a pre-registered rule picks $k$ on held-out episodes (mainly the largest $k$ at which round-to-nearest, det, loses at least 20 points), and we test $k-1$, $k$ and $k+1$. We ask why det hurts planning more than its one-step error suggests, and whether changing only the rounding repairs it.

### *B. Novelty*

(1) *Measuring exploitation.* The optimizer-exploitation index (OEI), the model-cost error of the executed plan minus that of random first-iteration candidates, splits exactly into a selection term and a drift term. (2) *A two-axis control*: stochastic variants with the same per-entry error that differ only in sharing across candidates and persistence across iterations. (3) *A training-free fix at the same bit-width* (sh-iter). We found no prior use of dither or stochastic rounding for low-bit world-model or VLA inference; the closest study [5] splits candidate-common and action-relative error with oracle repairs, without an iteration axis, a compression error or a deployable fix. These principles (the optimizer's curse [6], common random numbers, stochastic rounding [7]) are classical; their application to compressed world models, the two-axis separation and the effect size are new. (4) *Internal pre-registration*: hypotheses, thresholds and code were frozen in timestamped documents before each stage; failed hypotheses are reported (Table 1).

### *C. Algorithms*

**Rounding variants.** With grid step $\Delta$, det maps $x$ to $\Delta\mathrm{round}(x/\Delta)$; dither maps it to $\Delta\lfloor x/\Delta + u \rfloor$ with $u \sim U[0,1)$, which is unbiased. The stochastic variants differ only in how $u$ is drawn: per candidate (ind); shared by all candidates and redrawn at every forward call, hence fresh at every planner iteration (sh-iter, the proposed fix); shared and fixed within a control step (sh-step) or an episode (sh-ep; the two coincide in open-loop CEM). Other planner settings are the official defaults (MPPI: 64 elites; CEM: top 30, 10 iterations on Wall and PointMaze). **Exploitation.** With $e(a)$ the

cost error of candidate $a$ relative to the unquantized model, OEI $= e(a_{\text{exec}}) - \bar{e}_{\text{first}} =$ selection + drift, where the drift is the change of the population-mean error from the first to the last iteration. A negative drift means the planner moved toward candidates that the error favors. **Theory.** Fresh unbiased noise has zero expected drift, whereas a persistent field pulls the population to the minimizer of the smoothed model cost (a sample-average-approximation bias). The derived selection and drift terms matched 922k simulated MPPI and CEM runs (fitted slopes 0.92–1.05).

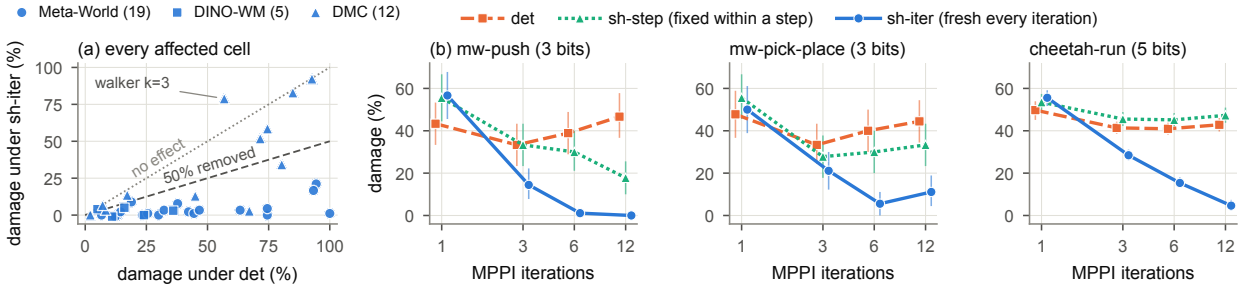



Fig. 1. (a) Damage under sh-iter vs. det in every affected cell (success-rate drop; relative return drop on DMC); below the dashed line sh-iter removes at least half. (b) Damage vs. MPPI iterations (TD-MPC2, 90 paired episodes, 95% CIs).

### *D. Experiments*

**Setup.** TD-MPC2: official 5M checkpoints (3 seeds), 8 Meta-World tasks (success) and 4 DMC tasks (return), 36 task–bit cells, 90 paired episodes per variant. DINO-WM: official PushT, Wall and PointMaze models, 9 cells, 100 paired episodes. All variants are paired with the unquantized baseline (same seeds and planner noise), and damage $D$ is the drop from it (relative return on DMC). A det drop counts as real if it survives per-task Holm correction (paired bootstrap, 95%); such cells are called affected (31 TD-MPC2 and 5 DINO-WM cells). **Phenomenon and fix** (Fig. 1a). det costs up to 100 points (mw-button-press, 2 bits). In 29 of these 36 affected cells its one-step error is no larger than that of sh-iter (mw-pick-place, 3 bits: 0.22 vs. 0.39, yet 37.8 vs. 7.8 points lost). sh-iter removes at least half of the damage in all 19 affected Meta-World cells (median 95%) and in 4 of 5 DINO-WM cells, and it is never worse than det in any affected Meta-World or DINO-WM cell. **Mechanism** (Fig. 1b). Under det the population-mean error drifts favorably in all four mechanism tasks at their primary bit-widths; sh-iter keeps at most 24% of that drift, while persistent dither in TD-MPC2 keeps 70–130%. The benefit needs iterations: with one iteration sh-iter is no better than det, but at 12 iterations its damage is 0–11% while that of det stays at 43–47%. With the elites fixed, exploitation and damage grow with the candidate count (3 of 4 tasks), and the damage of fixed dither shrinks as the number of independent offsets it draws (its table size) grows (5 of 6 tasks). **DMC.** The persistent share of det's favorable error (drift over OEI) is 0.33–0.52 on DMC versus 0.54–0.73 on Meta-World, and across the 12 TD-MPC2 tasks it predicts how much sh-iter helps (Spearman 0.83; a test fixed after stage 1 and run on new seeds). On DMC sh-iter removes half of the damage in only 5 of 12 cells and is harmful on walker at 3 bits (78.9% vs. 56.9% of return lost).

| Hypothesis and pre-registered test | Outcome |
|---|---|
| H1: det one-step error $\leq$ sh-iter in some affected cell | holds in both families |
| H3a: sh-iter removes $\geq$50% in $\geq$75% of affected cells; none $>$5 pts worse | Meta-World 19/19, DINO-WM 4/5; DMC fails |
| DMC rule (added after stage 1): Spearman of persistent share with gain $\geq 0.5$ | holds: 0.83 over 12 tasks |
| H2a′ (added after stage 1): det drift $< 0$, drift ratio sh-iter/det $< 0.5$ | holds: 4/4 tasks (ratio $\leq$0.24) |
| H2c: Spearman of $\lvert\text{OEI}_{\text{det}}\rvert$ and $D_{\text{det}}$ with candidate count $\geq 0.8$ | holds: 3/4 tasks |
| H4a: Spearman of fixed-dither damage with table size $\leq -0.8$ | holds: 5/6 tasks (three exactly at $-0.8$) |
| H3b: sh-iter beats det on top of the best PTQ in most 3–4-bit cells | TD-MPC2 holds (10/19); DINO-WM fails (1/3) |
| H5: dither latency overhead $\leq$10%; one Pareto-dominating setting | holds weakly (max 6.9%); no speed-up over fp32 |
| H2a, H2b (original OEI tests); H4b (near-tie density vs. sharing gain) | fail (claims narrowed) |

Table 1. Pre-registered hypotheses and outcomes. Mechanism tasks: mw-push, mw-pick-place, cheetah-run, PushT.

**References.** [1] N. Hansen, H. Su, X. Wang, TD-MPC2, ICLR, 2024. [2] G. Zhou et al., DINO-WM, arXiv:2411.04983, 2024. [3] Z. Fu et al., An empirical study of world model quantization, arXiv:2602.02110, 2026. [4] J. Xu et al., Predict before you deploy (PreDE), arXiv:2609.19441, 2026. [5] R. Min et al., When low prediction error misleads planning, arXiv:2610.05550, 2026. [6] J. E. Smith, R. L. Winkler, The optimizer's curse, Management Science, 2006. [7] S. Gupta et al., Deep learning with limited numerical precision, ICML, 2015.